

Practice Sheet 1.1: Basics of Python Programming

Level: Absolute Beginner (Python Basics)

1. Student Total and Average Marks

Scenario:

A student has scored marks in three subjects. Write a Python program to calculate the **total** and **average** marks.

Input:

Marks in Subject 1, Subject 2, Subject 3

Output:

Total marks and average marks

☐ *Concepts:* Variables, arithmetic operators

2. Simple Interest Calculator

Scenario:

A bank wants to calculate simple interest for a customer based on principal, rate, and time.

Input:

Principal amount, rate of interest, time (in years)

Output:

Simple interest

☐ *Concepts:* Formula-based computation

3. Area of a Rectangle

Scenario:

Given the length and breadth of a rectangle, calculate its area.

Input:

Length, breadth

Output:

Area of the rectangle

☐ *Concepts:* Multiplication, variables

4. Employee Salary Calculation

Scenario:

An employee receives a basic salary. Calculate the **gross salary** by adding HRA (10%) and DA (5%) of the basic salary.

Input:

Basic salary

Output:

Gross salary

☐ *Concepts:* Percentage calculations

5. Temperature Conversion (Celsius to Fahrenheit)

Scenario:

Convert the given temperature from Celsius to Fahrenheit.

Input:

Temperature in Celsius

Output:

Temperature in Fahrenheit

☐ *Concepts:* Mathematical expressions

6. Bill Amount Calculation

Scenario:

A customer purchases two items. Calculate the **total bill amount** including a fixed GST of 18%.

Input:

Price of item 1, price of item 2

Output:

Final bill amount

☐ *Concepts:* Addition, percentage, variables

7. Square and Cube of a Number**Scenario:**

Given a number, calculate its square and cube.

Input:

A number

Output:

Square and cube of the number

☐ *Concepts:* Exponents

8. Convert Minutes to Hours**Scenario:**

Convert the given number of minutes into hours.

Input:

Minutes

Output:

Hours

☐ *Concepts:* Division, float values

9. Average of Two Numbers**Scenario:**

Find the average of two numbers entered by the user.

Input:

Two numbers

Output:

Average value

□ *Concepts:* Arithmetic operations

10. Currency Conversion

Scenario:

Convert the given amount in Indian Rupees to US Dollars assuming
 $1 \text{ USD} = 83 \text{ INR}$.

Input:

Amount in INR

Output:

Equivalent amount in USD

□ *Concepts:* Division, constants